

Reducing artifacts during human intracranial electrical stimulation by separating current return from recording ground

Dear Editor

Direct electrical stimulation is an essential tool for treating neurological and psychiatric disorders, investigating brain function, and developing brain-computer interfaces [1–6]. To understand the effects of electrical stimulation, it is often necessary to record neural activity before, during, and after a pulse train is delivered. However, recording useful neural data during and immediately after stimulation can be difficult because the artifacts induced by stimulation are several orders of magnitude larger than the evoked neural response. To deliver electrical stimulation to an electrode, a current return path must be established, and the size and impact of artifacts vary with the path of current return selected. For example, bipolar stimulation uses a local current return and typically has a smaller artifact, whereas monopolar stimulation uses a distant return, commonly the recording ground, and typically has a larger artifact [7]. For monopolar stimulation, we demonstrate that the simple practice of separating current return from recording ground consistently and significantly reduces stimulation artifacts.

Similar to electrical stimulation requiring a return path, an amplifier recording neural activity from an electrode requires a reference point. Commercially available integrated electrophysiology systems that provide neural recording and electrical stimulation can be classified into one of two categories. One type of system uses a *shared* configuration (Fig. 1A) in which the current return for the stimulator is wired to the recording ground (often an electrode embedded in cortical bone or white matter). The other type of system uses a *separate* configuration (Fig. 1B), in which one contact, such as a conductive pad placed on the patient's thigh, is used as the current return for stimulation, and a separate contact (in bone or white matter) is used as the recording ground. Both shared and separate configurations are in common use: devices from Blackrock Neurotech and Ripple Neuro use a shared configuration while devices from Natus and Tucker-Davis use a separate configuration.

To compare artifacts, neural signals were recorded while a 1 s pulse train was delivered to an intracranial electrode using both configurations (see SOM for detailed *Methods*). For a representative recording electrode (e4), artifacts were visibly larger for the shared configuration compared to the separate configuration (Fig. 1A vs. 1B). To quantify this difference, we examined the artifact at four different time windows (artifact definitions shown in SOM Fig. 1). The RMS difference between the shared and separate configurations were calculated and a statistical comparison performed.

The shortest time scale was that of individual pulses; Fig. 1C shows the pulse artifact. In the shared configuration, the recorded voltage rapidly exceeds the maximum measurable voltage of the recording amplifier and remains at the upper or lower extreme for the duration of

the biphasic pulse. The pulse artifact caused the amplifier to saturate, or “rail”, meaning that no neural signal could be recorded for the duration the signal amplitude exceeded the amplifier limit (amplifier saturation) leading to data loss. When a separate current return path was used, the amplitude of the artifact was significantly smaller (see SOM Table 1 for values and details of statistical comparison) and there was no data loss from amplifier saturation.

The next artifact considered occurred between pulses; Fig. 1D shows the interpulse artifact. In the shared configuration, the signal showed a clear deflection after the end of the pulse, followed by a gradual return toward baseline during the interpulse interval. This artifact was not observed in the separate configuration. A third artifact occurred on the time scale of the entire pulse train (Fig. 1E). In the shared configuration, there was a large initial deflection that increased in amplitude over the course of the pulse train that was significantly larger than in the separate configuration. A fourth artifact occurred following the end of the pulse train (Fig. 1F). In the shared but not the separate configuration, a brief negative offset was followed by a positive rebound that decayed over several seconds.

The analyses were repeated across all electrodes and for four additional stimulation protocols that differed in pulse width and pulse train duration (Supplementary Table 2). Across stimulation protocols, the shared configuration consistently resulted in significantly larger artifacts and longer periods of amplifier saturation compared to the separate configuration.

Having demonstrated significantly worse artifacts for the shared configuration than the separate configuration, we examined the impact on the ability to record neural signals. For two electrodes with stimulation-induced neural activity, the activity could be observed in the separate but not the shared configuration (SOM Fig. 2). The inability to record neural signals in the shared configuration was due to the long duration of the post-train artifact.

These results highlight the importance of using a separate, rather than a shared configuration, during brain stimulation. For systems such as the Blackrock CereStim that use a shared configuration, it is possible to modify the manufacturer-supplied cable to separate the stimulation current return and the recording ground [8], as in the present experiments.

In the shared configuration, there is direct contamination of all recorded electrodes by the stimulation pulses within the amplifier. In contrast, in the separate configuration, recording artifacts primarily result from bulk current spread. This means that electrodes closest to the stimulating electrode tend to suffer the largest artifacts, with more distal electrode having smaller artifacts.

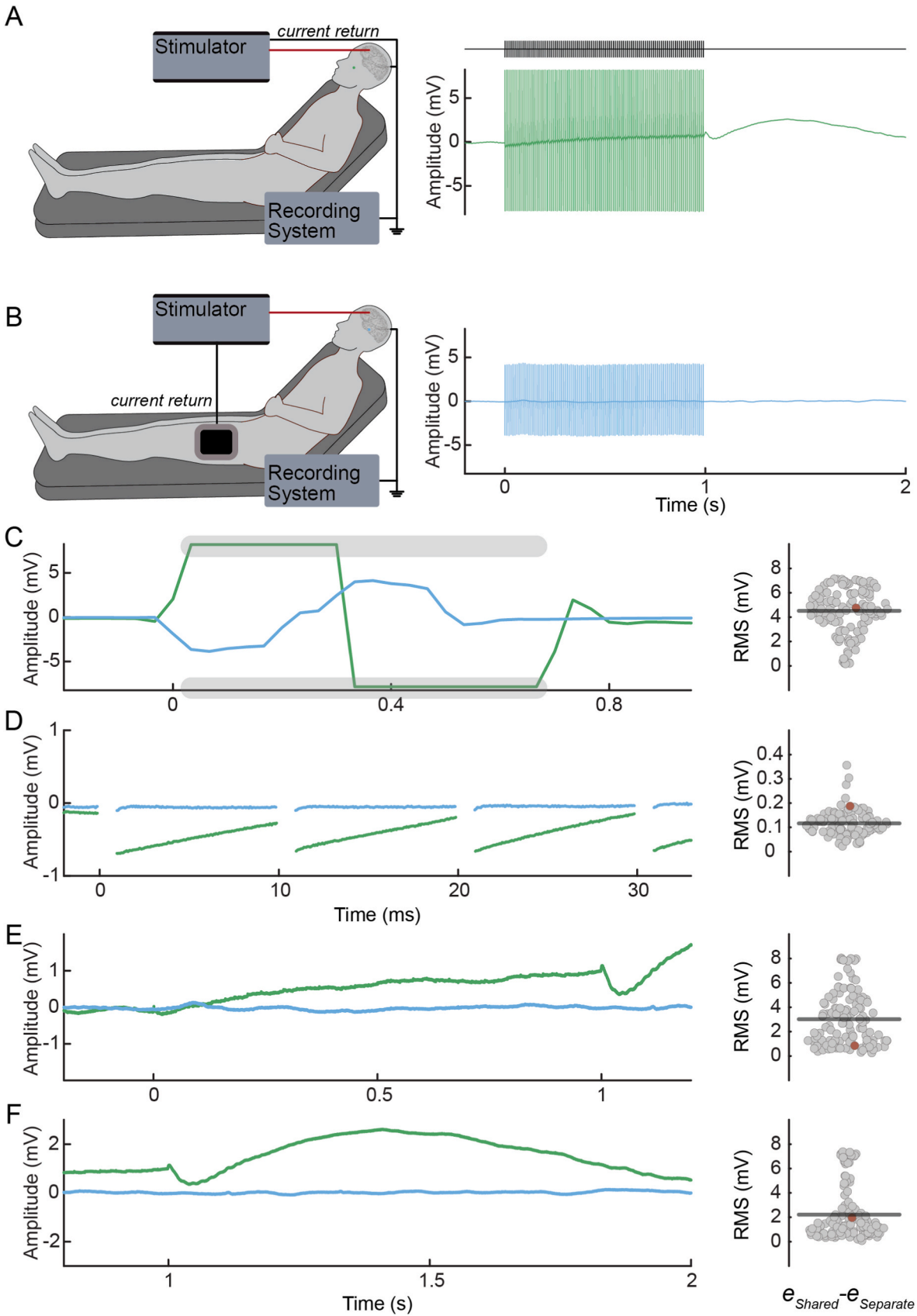
The shared configuration poses another potential issue for monopolar stimulation. In the epilepsy monitoring unit, an electrode in white

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Fig. 1. Stimulation artifacts for neural recordings collected with shared or separate current returns. Examples are presented for a recording collected during delivery of a 1 s pulse train (200 μ s, cathodal-first, biphasic, 100 μ s inter-phase gap) delivered at 100 Hz. **A. Left:** In the “shared” configuration, the neural stimulator shares the current return with the recording system ground (an sEEG electrode contact within the skull.) **Right:** The signal recorded from a representative electrode, *e4*, during the pulse train with a shared current return. **B. Left:** In the “separate” configuration, a neurosurgical grounding pad placed on the patient’s thigh was used as a current return. **Right:** The signal recorded from *e4* during stimulation with a separate current return. The recorded signal was decomposed into artifacts that occurred at four different time scales. **C.** Pulse artifact recorded on electrode *e4*. In the shared configuration (green line), the recording system reached the amplifier limit and clipped the signal at ± 8 mV (thick gray line). In the separate configuration (blue line), the artifact was smaller in amplitude. The root mean square (RMS) of the signal was calculated for each configuration (averaged across stimulation pulses). In the right plot, each symbol represents one electrode and shows the difference in RMS for each electrode (shared - separate). The red symbol shows electrode *e4*. **D.** The artifact that occurs between pulses (interpulse artifact) on electrode *e4*; the single pulse and pulse train artifacts were removed from the signal to isolate the interpulse artifact. Between each stimulation pulse (shown with black ticks on x-axis) an artifact is visible in the shared configuration (green line) but not the separate configuration (blue line). For each recorded electrode, the RMS was larger when the recording ground was shared with the current return. **E.** The artifact that occurred across the pulse train (pulse train artifact) for electrode *e4*; the single pulse and interpulse artifacts were removed from the signal to isolate the artifact. Across the pulse train, an artifact is visible in the shared configuration (green line) but not the separate configuration (blue line). For each recorded electrode, the RMS was larger with a shared configuration. **F.** The final artifact (post-train artifact) occurred after the pulse train terminated at $t = 1$ sec in electrode *e4*. The artifact was quantified by calculating the RMS in the 1 sec following the pulse train for each configuration.

matter is routinely selected as recording ground [9]. In the shared configuration, the recording ground also serves as the current return path, which may inadvertently activate white matter with potential adverse consequences, such as inducing a seizure. While this may be done intentionally with bipolar stimulation for clinical purposes, it is important to avoid accidental or unintended activation during research.

Stimulation artifact can distort or obscure relevant electrophysiological responses across multiple time scales. If the amplifier is fully saturated, the data cannot be recovered, leading to data loss. Some of the neural effects of stimulation may only occur at the beginning of the pulse train or cease immediately after stimulation. Evoked potentials, for example, can occur within the first 10 ms following electrical stimulation [10], and slow amplifier recovery from saturation means this early response can be missed entirely. While using a separate configuration does not eliminate all artifacts, it does reduce artifact magnitude and duration, which helps ensure that useable neural data can be collected during and immediately after stimulation.

CRedit authorship contribution statement

Denise Oswalt: Conceptualization, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Michael S. Beauchamp:** Funding acquisition, Writing – original draft, Writing – review & editing. **Han-Chiao Isaac Chen:** Investigation, Resources. **Daniel Yoshor:** Funding acquisition, Investigation, Project administration, Supervision.

Declaration of generative AI and AI-assisted technologies in the writing process

No AI was used in the construction or editing of this manuscript.

Declaration of competing interest

Denise Oswalt.
I have nothing to declare.
Michael Beauchamp.
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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.brs.2025.10.017>.

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